

ASA Data Quest

**DATA SCIENCE AND SUSTANABLE
DEVELOPMENT: OPTIMIZING ECONOMIC, SOCIAL,
AND ENVIRONMENTAL DIMENSIONS**

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OVERVIEW

● Data Source

World Development Indicators (WDI), covering annual country-level data from 1960 to 2022.

Variables

- **GDP** – economic dimension
- **Education rate** – social dimension
- **Life expectancy** – human development dimension
- **Health expenditure** – welfare indicator
- **CO₂ emissions** – environmental dimension

● Data Preparation

- Missing observations were addressed.
- Units were standardized.
- Outliers were checked and controlled.

● Statistical Diagnostics

Autocorrelation (Durbin–Watson test): Independence of errors confirmed.

Multicollinearity (Variance Inflation Factor, VIF): no evidence of high collinearity among variables.

Normality (Shapiro–Wilk test): residuals examined for normal distribution.

● Research Objective

The study aims to analyze the relationship between GDP and selected economic, social, and environmental indicators, and to interpret these linkages within the framework of a multidimensional sustainable development model.

ANALYST & RESULTS

ANALYST

Missing data control and data cleaning: Incomplete or extreme observations were removed from the dataset.

Variable standardization: Indicators were normalized to account for differences in measurement scales.

Correlation analysis (Pearson): Linear associations between GDP and the other variables were assessed.

Trend analysis (regression trends): The linear effects of GDP on education and life expectancy were visualized.

Reliability check: Internal relationships among variables were validated using the correlation matrix.

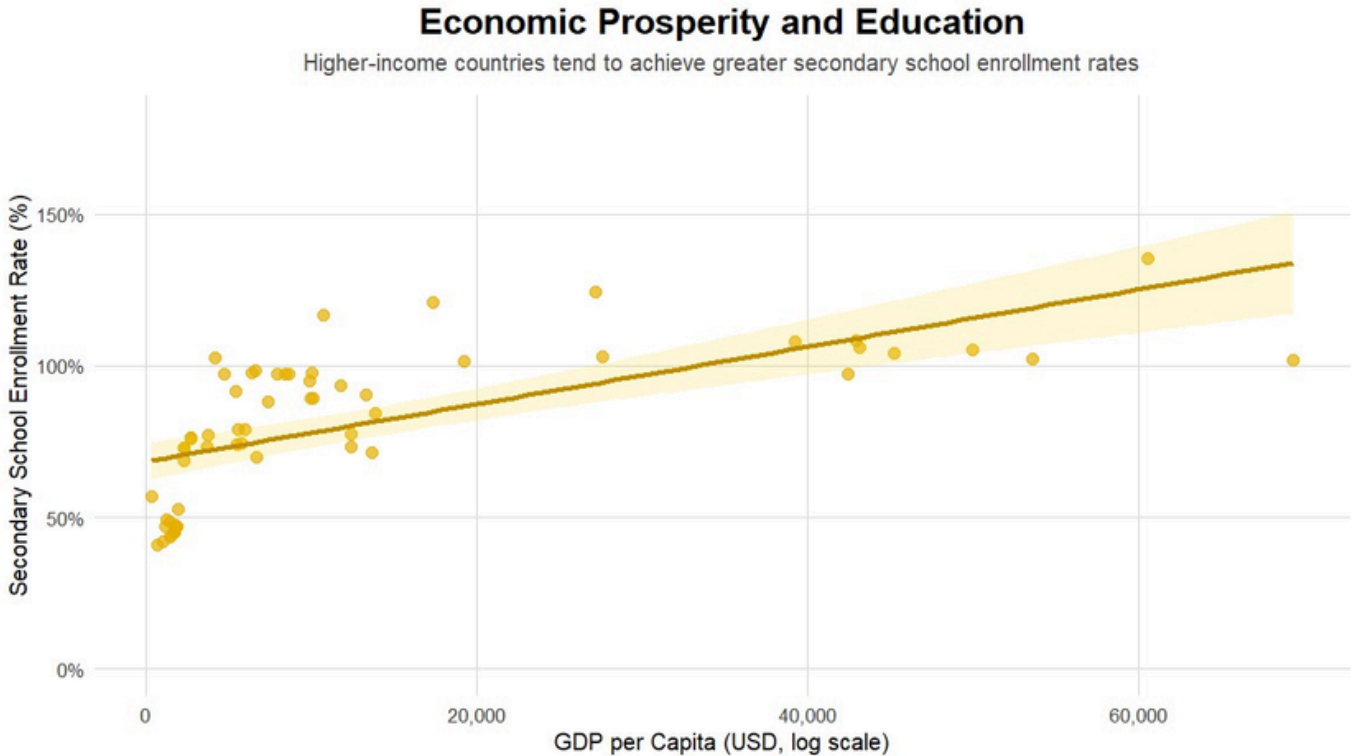
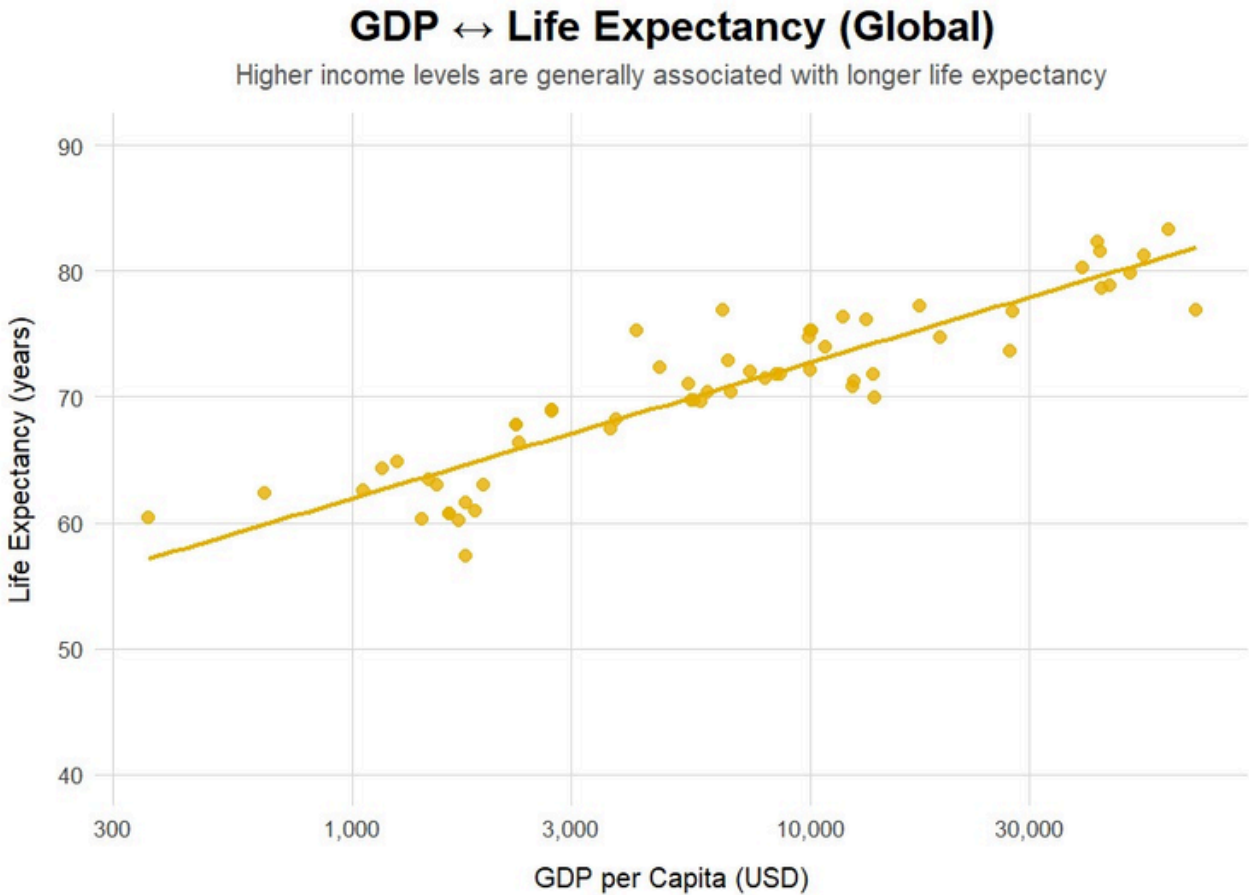
RESULTS

A strong positive correlation ($r \approx 0.8-0.9$) was identified between GDP, education, and life expectancy.

Increases in education levels are associated with higher GDP, suggesting that economic growth reinforces social development.

Life expectancy also shows a robust linear relationship with GDP.

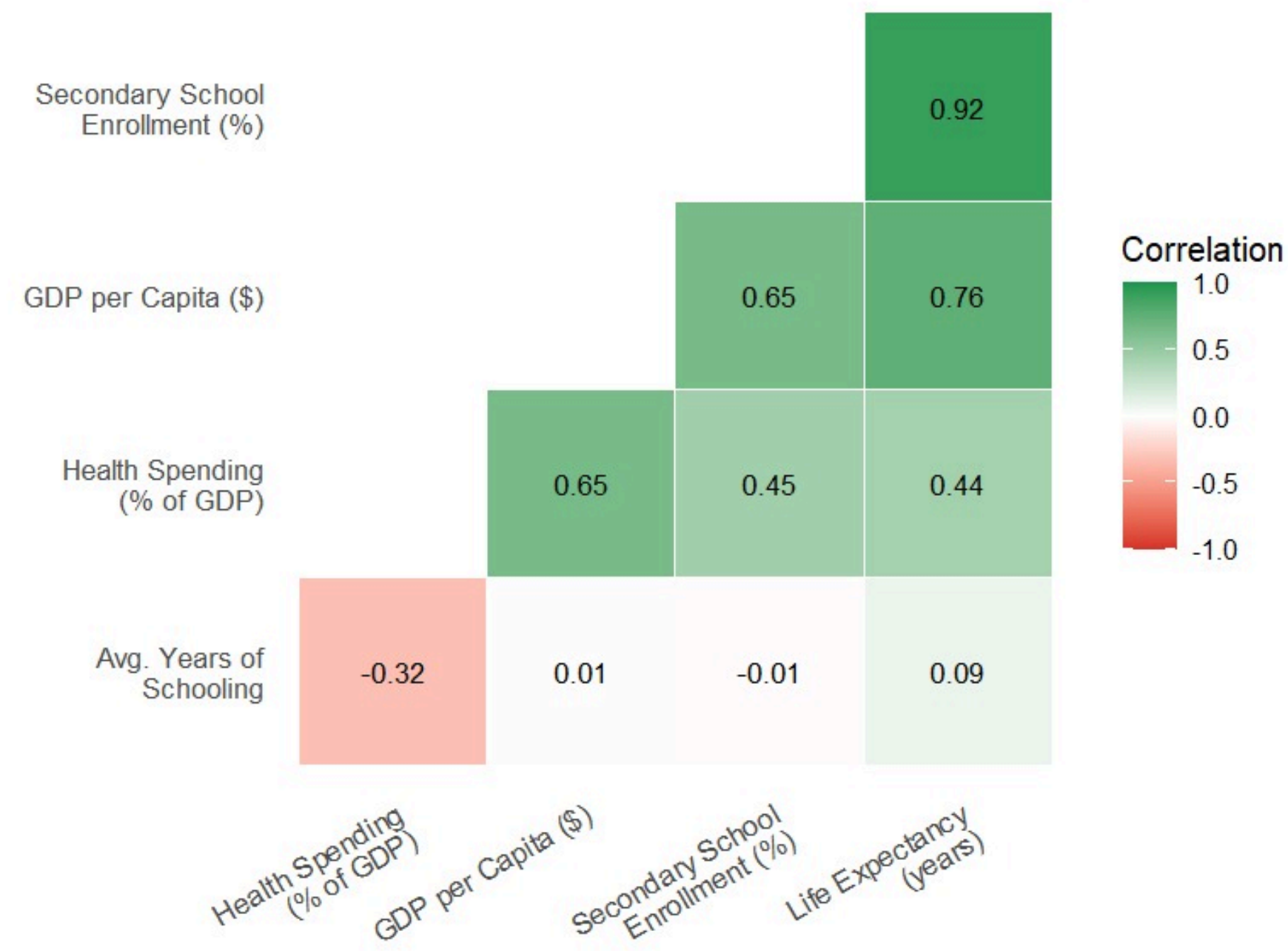
Health expenditure plays a mediating role between GDP, education, and life expectancy.



ANALYST & RESULTS

Socioeconomic & Health Indicators Correlation Matrix

GDP, Education, Life Expectancy, and Health Spending



This correlation matrix reveals the relationships between key economic and social indicators.

- Notably, the strong positive correlations between GDP, education, and life expectancy highlight that economic prosperity progresses in parallel with social development.
- The correlation of **0.92** between education and life expectancy emphasizes that investment in education directly translates into improvements in quality of life.
- Furthermore, the positive relationship between health expenditures and **GDP (0.65)** indicates that rising income levels lead to greater allocation of resources toward healthcare systems.

HAPPINESS AND GREEN GROWTH

We chose not to limit our analysis exclusively to economic and social indicators.
In addition, we integrated data from the ***World Happiness Report*** into our study.

Our objective was to examine the relationship between countries' economic prosperity (GDP), happiness scores, and environmental performance (CO₂ emissions, renewable energy usage).

According to the *World Happiness Report*, the happiest countries are typically high-income, industrially advanced nations.

However, what surprised us was this finding:

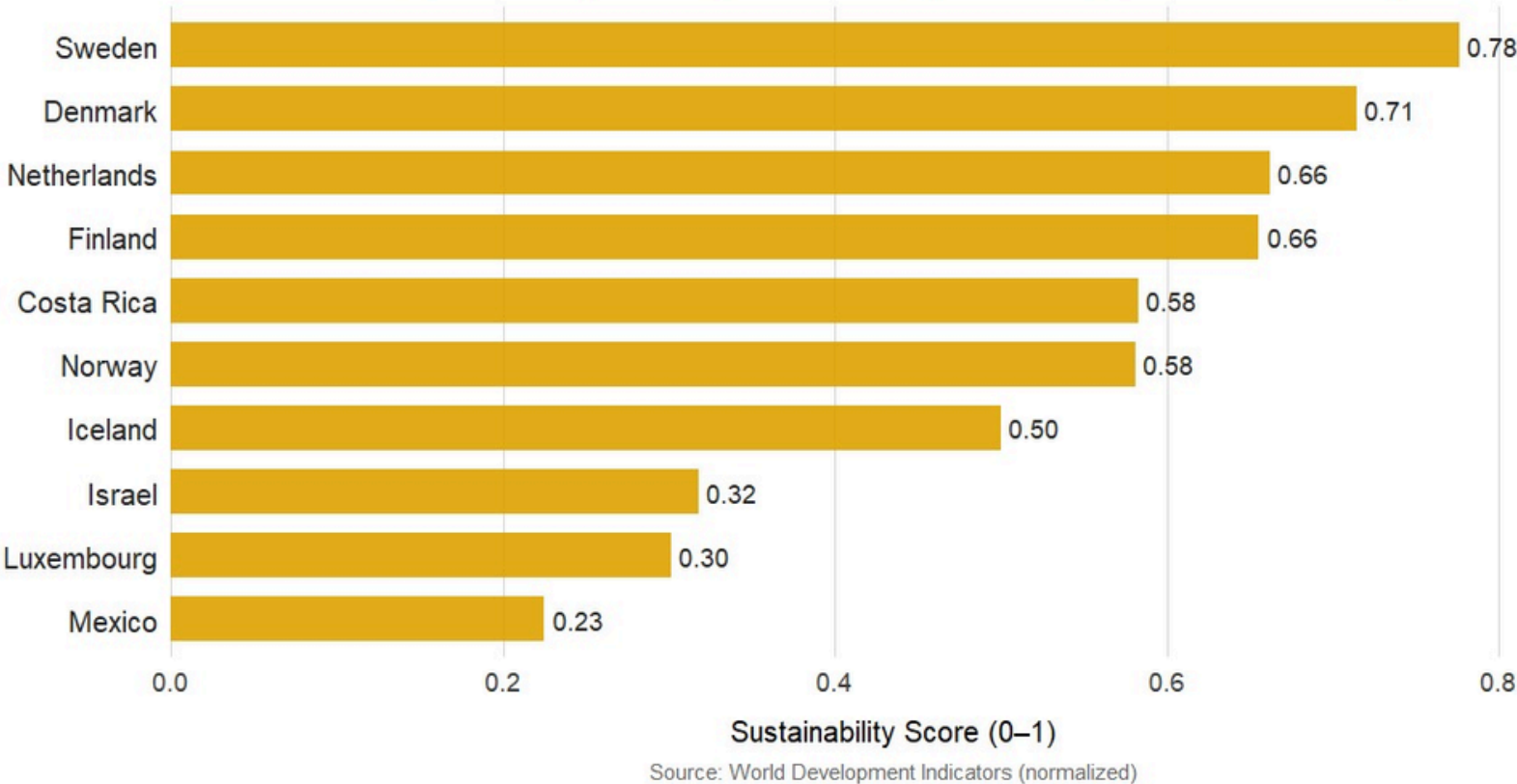
The world's most industrialized economies are also among the countries with the **lowest carbon emissions**.

In these nations, economic growth is achieved **without exhausting nature**—instead, it is sustained through renewable energy adoption and clean production policies.

In conclusion, we observed that prosperity is not defined by income alone; rather, it emerges from the balance **between wealth, happiness, environmental responsibility, and sustainability**.

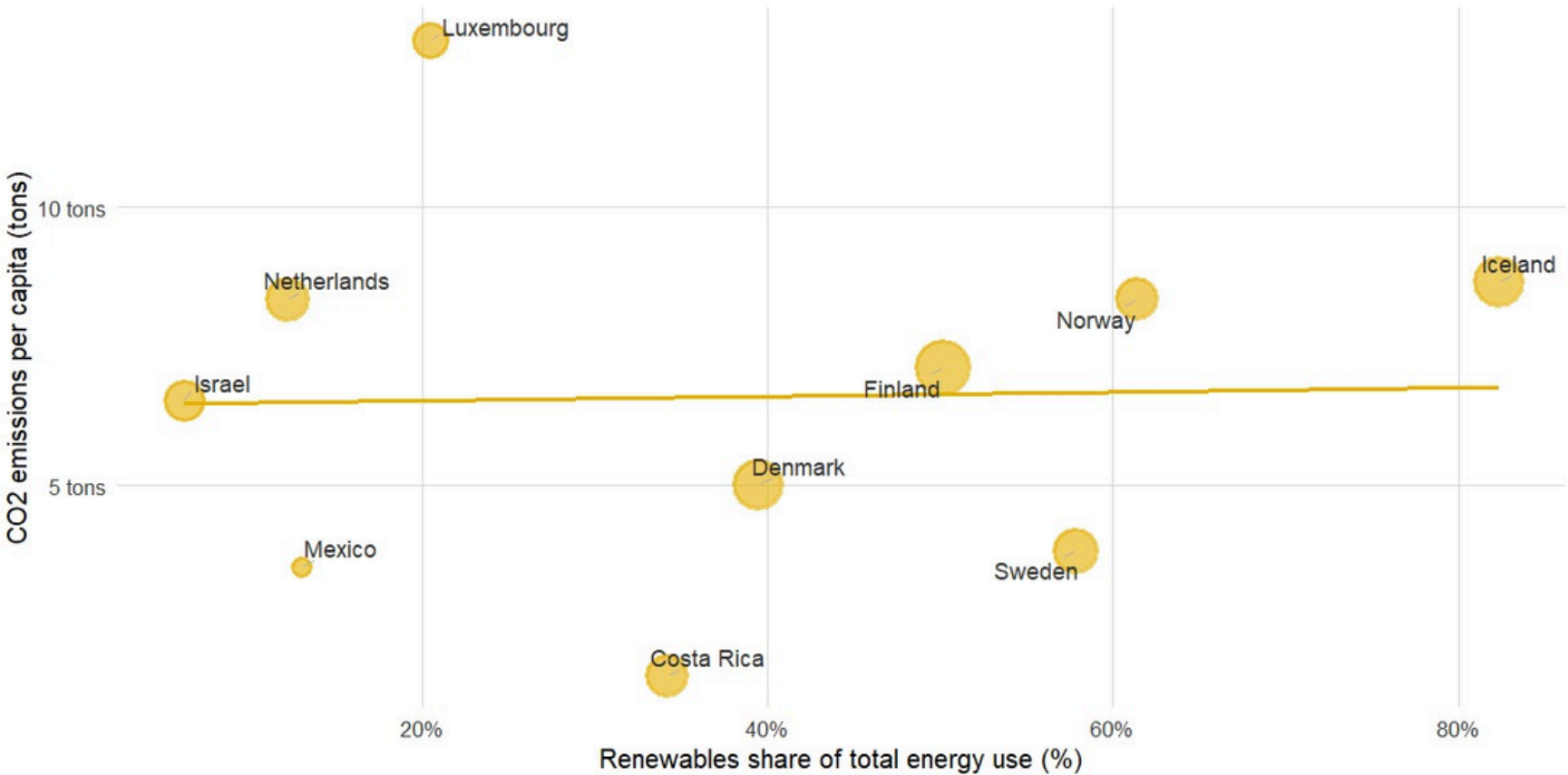
Top 10 Happiest Countries – Sustainability Score

Higher score implies stronger sustainability performance (environmental + social)



Renewables vs CO2 — bubble size indicates Happiness

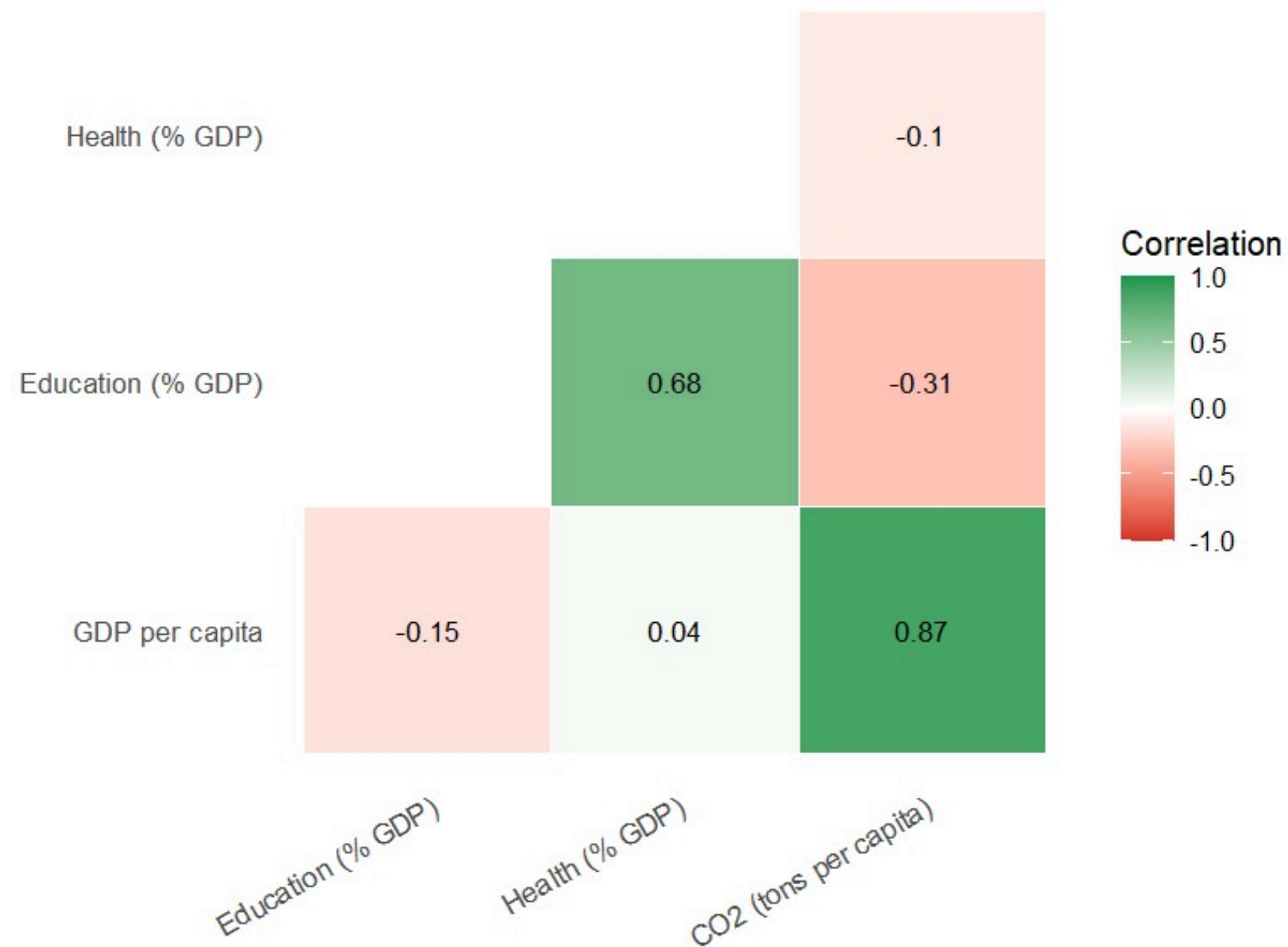
Countries with higher renewable shares tend to show lower CO₂ emissions and higher happiness levels



HAPPINESS AND GREEN GROWTH

Socioeconomic & Environmental Indicators Correlation Matrix

GDP per capita, Education Spending, Health Spending, CO2 Emissions



This correlation matrix illustrates the relationships between economic (GDP), social (education and health), and environmental (CO₂ emissions) indicators.

- The strong **positive correlation** between **GDP** and **CO₂ emissions (0.87)** demonstrates that economic growth is closely tied to energy consumption and carbon output.
- In contrast, the weak or negative correlations of education and health indicators with **CO₂ emissions** reveal that social development carries a much lower environmental cost compared to economic growth.
- These findings emphasize that **sustainable development** is not driven by economic growth alone, but also by socially inclusive progress that aligns with environmental responsibility.

OPTIMIZATION

Extended Contribution Model Purpose

In our study, we did not limit ourselves to analyzing existing data; rather, we aimed to create a forward-looking perspective by developing an **optimization** model.

The model **integrates** economic, social, and environmental indicators to quantitatively assess countries' positions in sustainable development.

Core Components

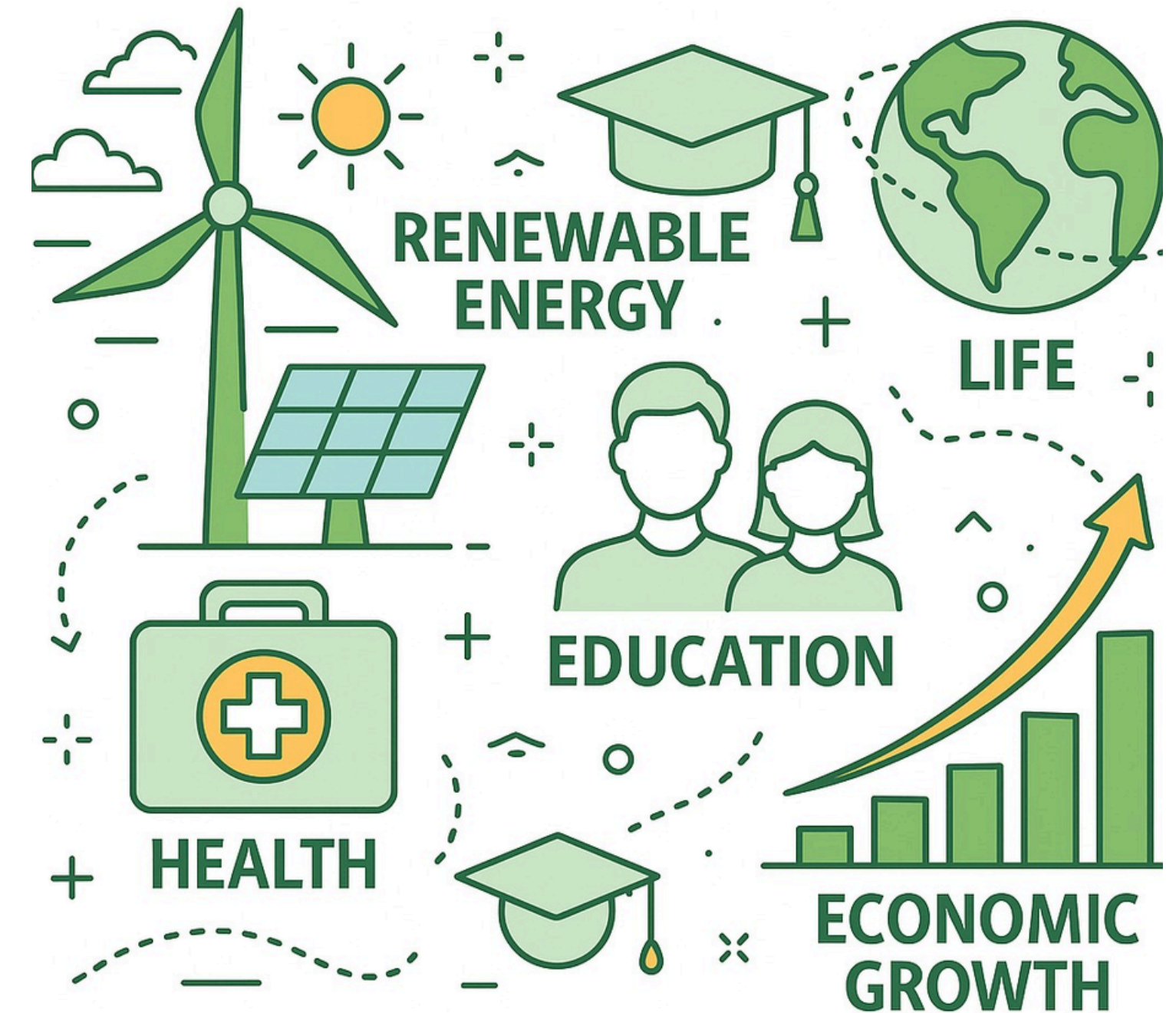
GDP growth → economic capacity
Education rate → social development
Health expenditure → human welfare
Carbon emissions (inverted) → environmental balance

Methodology

All indicators were normalized and equally weighted within the optimization framework, resulting in a **sustainability** score for each country.

Findings

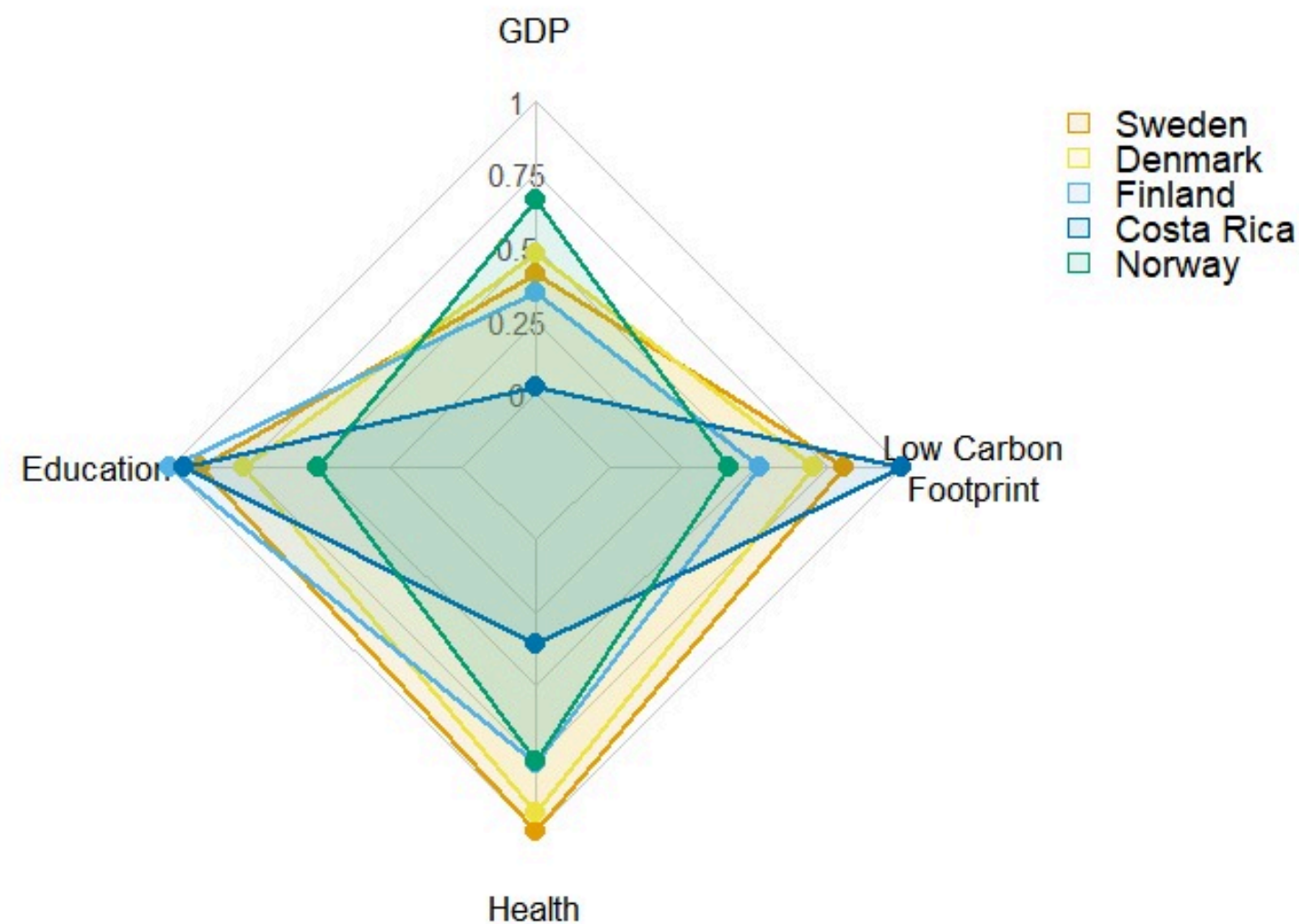
The results reveal that Scandinavian countries, along with **Costa Rica**, demonstrate the most balanced profiles in terms of economic strength, social progress, and environmental sustainability.



OPTIMIZATION

Sustainability Profile — Selected Countries

Scores are normalized to [0–1]. Higher is better on all axes.
CO₂ is inverted → cleaner energy = higher score.



The radar chart illustrates the multi-dimensional sustainability profiles of selected countries, based on our optimization model.

- Each axis represents a key component of **sustainable** development
- **Economic Capacity (GDP)**, Education, Health, and Low-Carbon Footprint (**CO₂ inverted**).
- The visualization shows that **Scandinavian** countries (Sweden, Denmark, Finland, Norway) maintain balanced and high-level performances across all dimensions, while Costa Rica stands out for its strong environmental performance despite its **relatively** lower GDP.

An aerial photograph of a lush green landscape, featuring a winding road, dense forests, and a small settlement with several buildings. The image is overlaid with a semi-transparent green filter. The text "Thank You!" is written in a large, white, cursive font across the center of the image.

Thank You!